Rank Where It Counts: Budgeting LoRA on GPT-2 Medium with *Depth-Weighted LoRA*, *AdaLoRA-Lite* & *MoLE*

CS 4782 · FINAL PROJECT
Justin Xiang · Eric Qiu
jx372@cornell.edu · sq225@cornell.edu

1 Why Care?

- Fine-tuning a 354M-param transformer used to mean storing a fresh 354M-param checkpoint **per task**. **LoRA** (Hu '21) trains tiny rank-r add-ons next to frozen weights instead. In Table 3, GPT-2 Medium LoRA reaches:

0.35M	**70.4**	**~0**
trainable params (~1000× less)	BLEU on E2E (beats full FT)	extra latency at inference

OUR QUESTION

LoRA spends the **same rank on every layer**. But layers don't all do the same thing — **so should the budget?** We try two ways to redistribute it (one by hand, one learned), plus a bonus: one router, three task LoRAs, one served model.

2 LoRA Reparameterization

$$h = W_0 x + (\alpha / r)\, B A x$$

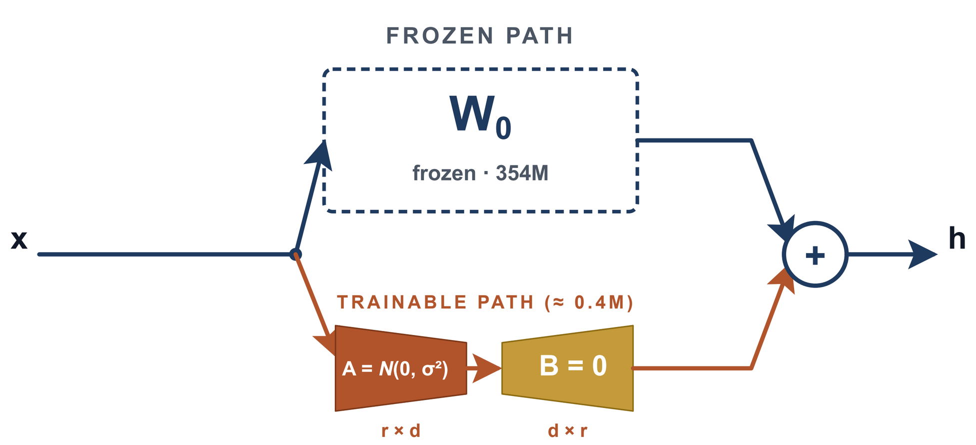

Fig. 1. Rank-r adapters parallel to frozen Q/V; $A \sim N(0,\sigma^2)$, B=0 at init.

3 Three Datasets, Same Idea

```
name : Blue Spice | type : coffee shop | rating : 5/5 |
near : Crowne Plaza Hotel
```

"Blue Spice is a coffee shop near Crowne Plaza Hotel with a 5-out-of-5 rating."

Model	GPT-2 Medium (354M, frozen)
Adapters on	attention **Q, V** only
Train	5 epochs, AdamW, fp16, 1×A100

- **E2E.** Restaurant attributes → description (42K pairs).
- **DART.** Open-domain triples (Wikipedia, knowledge graphs) → sentence.
- **WebNLG.** RDF triples → multi-sentence paragraph.
- **Runs:** 5 uniform $r \in \{1,2,4,8,16\}$ · 3 late-ramp · 3 r=4 ablations · 3 AdaLoRA-Lite $r_{max} \in \{6,8,16\}$ · all hit 192 active rank units.

QUESTION 1

If transformer layers specialise with depth, should later layers get more rank and early layers less?

Two ways to redistribute the same parameter budget: by hand (Card 4) or learned (Card 5).

4 Extension #1 — Depth-Weighted LoRA (Late-Ramp)

- Same 192-unit budget as uniform r=4, redistributed by **depth**: small/early, larger/late.

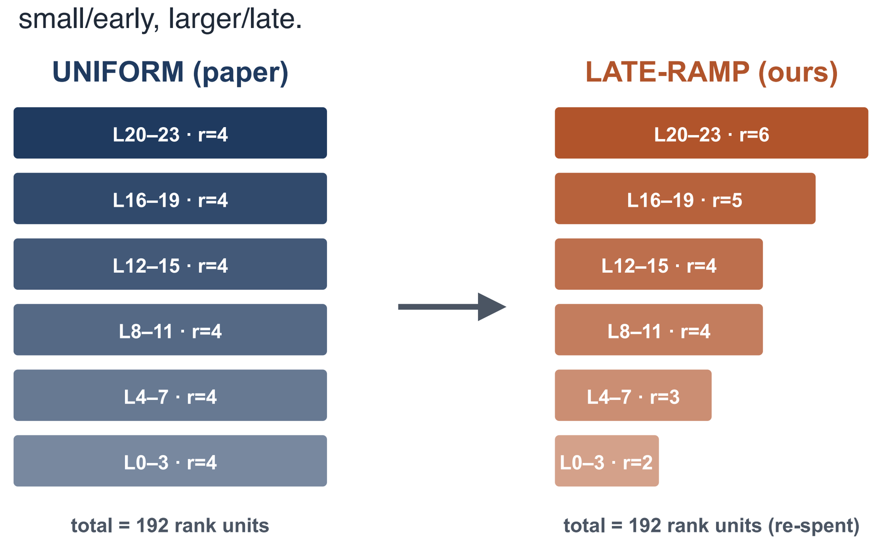

Fig. 2. Same total rank, redistributed by depth.

QUESTION 2

If we allow many rank directions but keep pruning by sensitivity, can we match a hand-designed depth profile at the same active budget?

Complements Q1: same 192-unit cap; here gates and pruning learn the profile instead of prescribing late-ramp.

5 Extension #2 — AdaLoRA-Lite

- Each rank direction gets a learnable gate s_i; let the model learn the profile.
- Every **500 steps**, rank gates by EMA($|s_i \cdot \partial L/\partial s_i|$) (β=0.85) and prune to **192 active units**.

$$\Delta W = (\alpha / r_{max}) \cdot B \cdot \mathrm{diag}(s) \cdot A$$

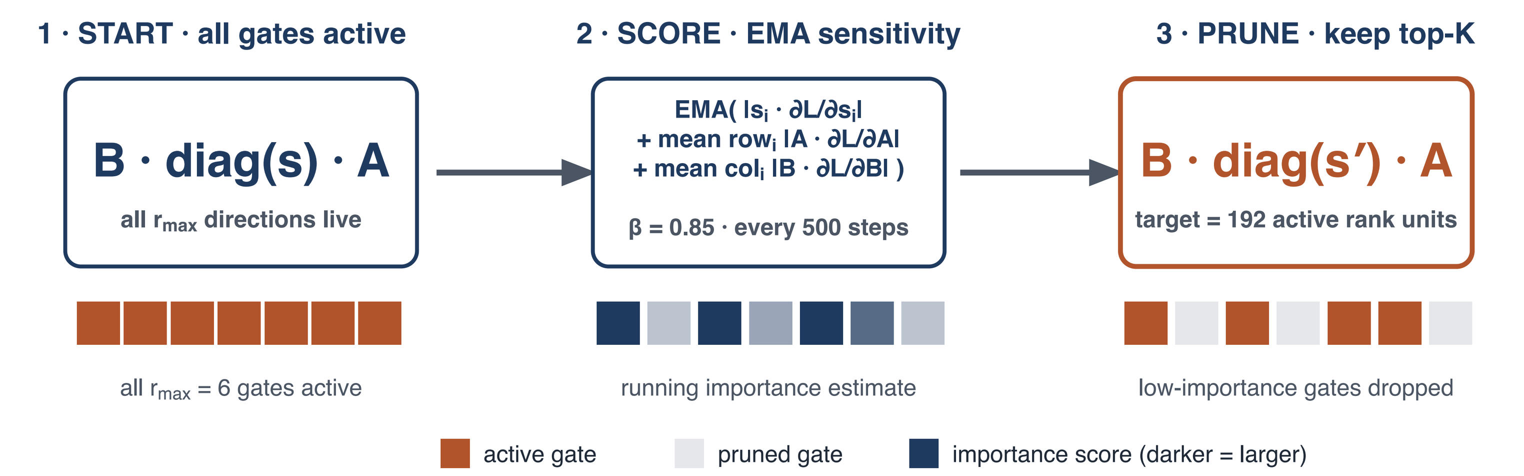

Fig. 3. Score gates, prune to budget along cubic schedule.

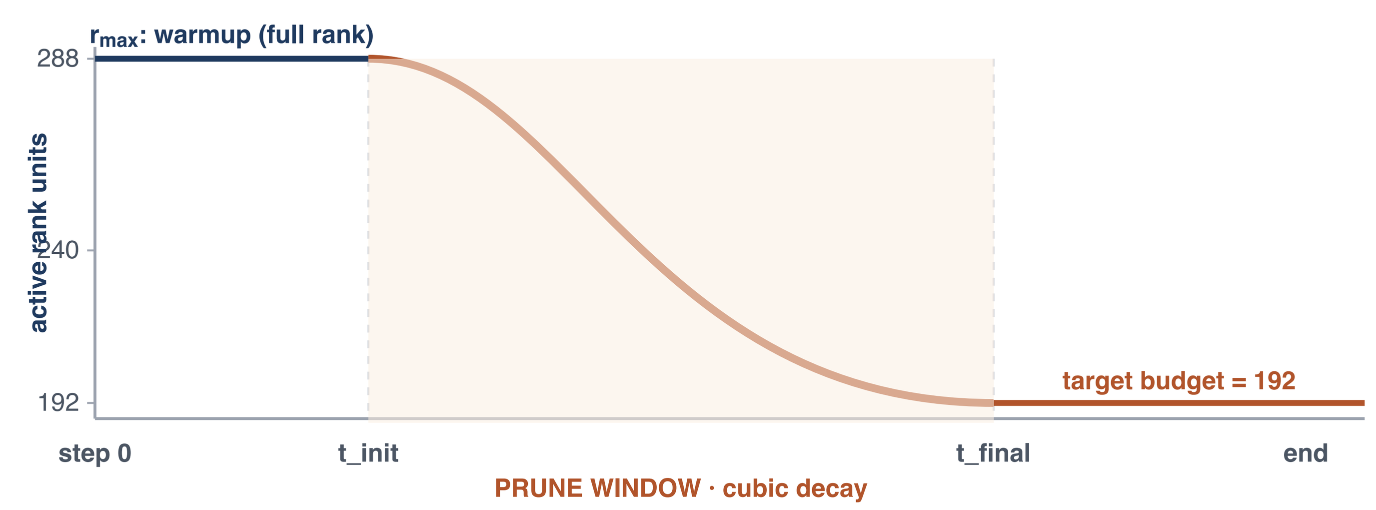

Fig. 4. Cubic decay from r_{max} to 192. Best: t_init=10K, t_final=21K of 26K.

Results

Want the full Q1 results? Scan to view extended ablations and per-task tables we omitted from the poster.

`github.com/justinlxiang/CS4782-final-project`

6 Where to Spend a Tight Budget?

r=4 projection ablations · + AdaLoRA-Lite r_{max}=6 (0.39M) · BLEU on E2E

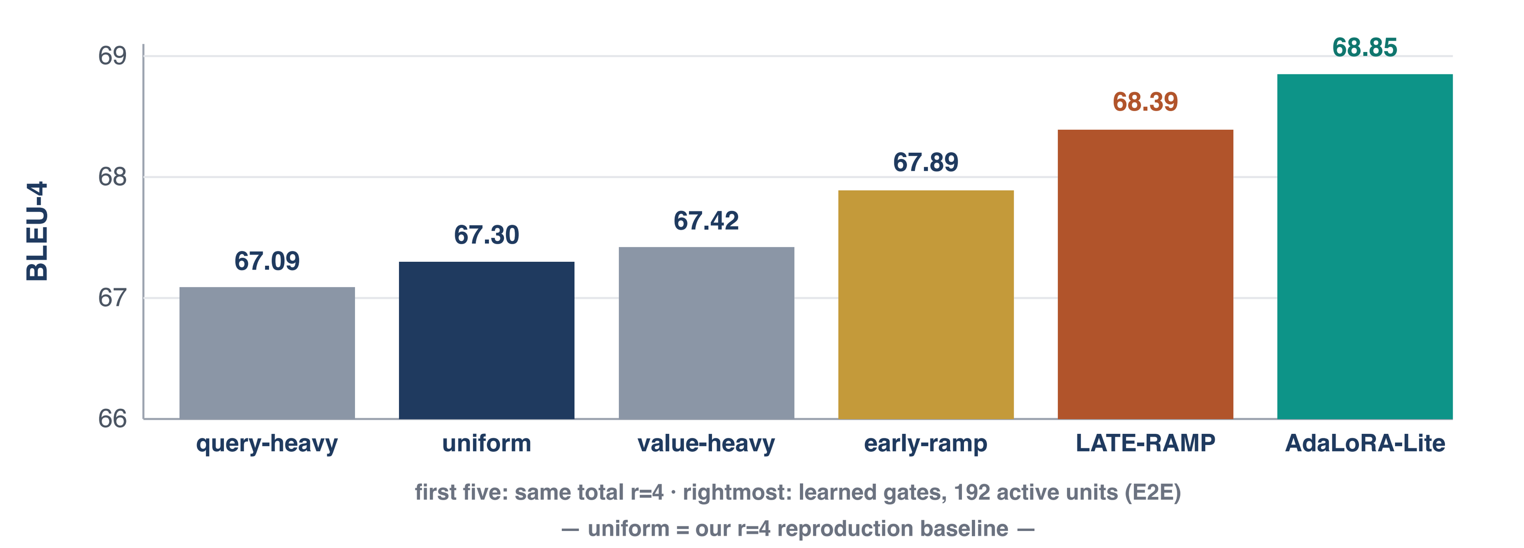

Fig. 5. Depth (late-ramp) beats projection ablations on the same r=4 budget; AdaLoRA-Lite r_{max}=6 is our strongest matched-budget run.

7 E2E Test Set — Official Metrics

Configuration	Params	BLEU	NIST	METEOR	ROUGE-L	CIDEr
★ *LoRA r=4 (Hu '21)*	*0.35M*	*70.40*	*8.85*	*46.80*	*71.80*	*2.53*
★ *Full fine-tuning (paper)*	*354.92M*	*68.20*	*8.62*	*46.20*	*71.00*	*2.47*
★ *AdapterL 11M (paper)*	*11.09M*	*68.90*	*8.71*	*46.10*	*71.30*	*2.47*
Uniform r=1	0.10M	66.78	8.51	45.02	68.94	2.36
Uniform r=2	0.20M	67.11	8.52	45.78	70.30	2.45
Uniform r=4 (ours)	0.39M	67.30	8.51	46.05	70.96	2.47
Uniform r=8	0.79M	67.97	8.59	46.23	70.94	2.50
Uniform r=16	1.57M	68.21	8.62	46.22	71.12	2.49
Late-ramp r=4	0.39M	68.39	8.63	46.51	71.29	2.49
Late-ramp r=8	0.79M	67.77	8.57	46.28	70.90	2.48
Late-ramp r=16	1.57M	68.15	8.62	46.38	71.24	2.49
AdaLoRA-Lite r_{max}=16	0.39M*	67.67	8.59	46.10	70.50	2.45
AdaLoRA-Lite r_{max}=8	0.39M*	68.34	8.64	46.30	71.00	2.49
AdaLoRA-Lite r_{max}=6 ★ best	**0.39M***	**68.85**	**8.69**	**46.40**	**71.42**	**2.51**

Fig. 6. Official E2E metrics. Bold = our best extension.

8 What AdaLoRA-Lite Learned

final active rank · per layer · r_{max}=6

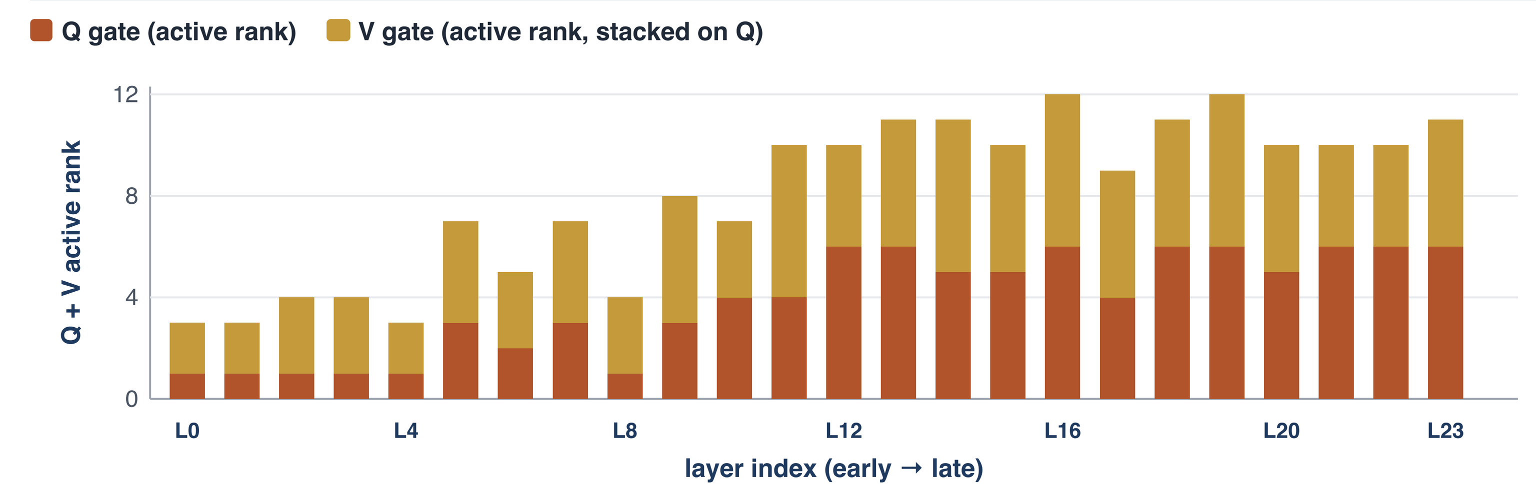

Fig. 7. Active rank settles into a late-layer-heavy profile.

9 AdaLoRA-Lite Rank Evolution

total active rank (Q + V) per layer · every 500 steps · r_{max}=6

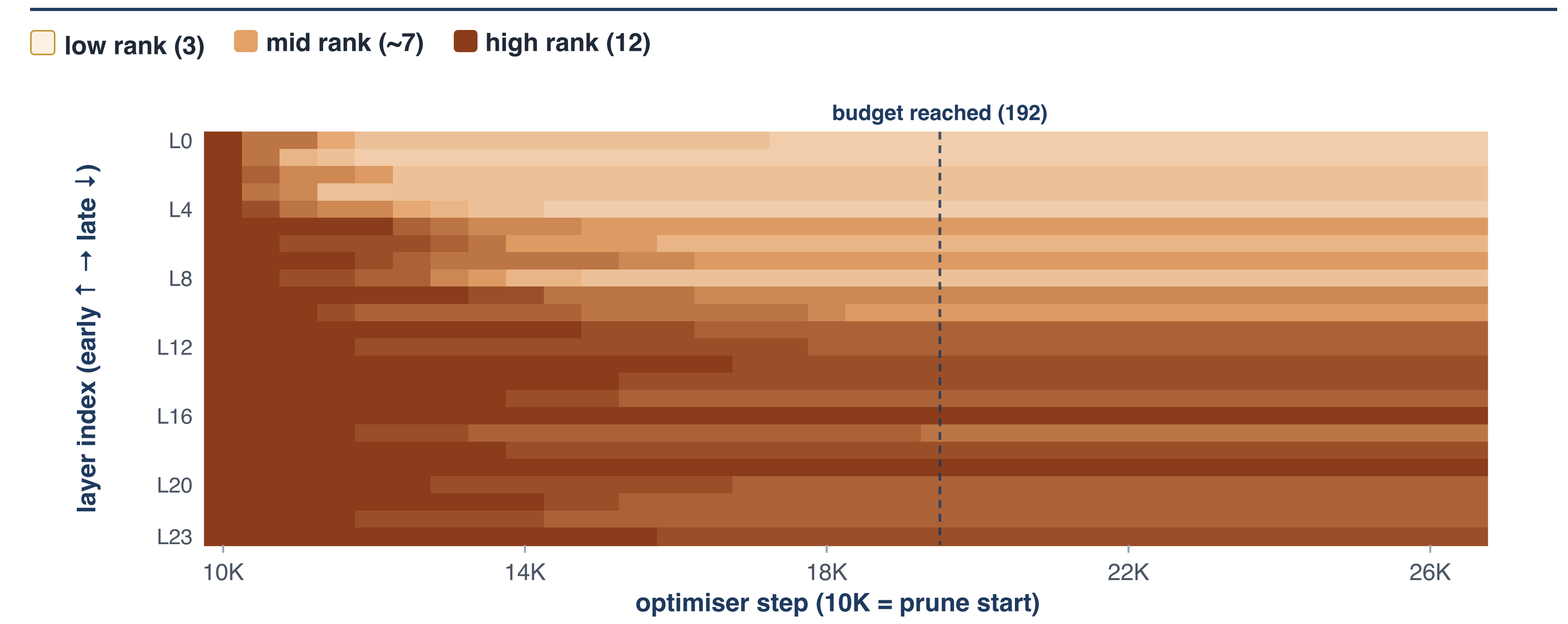

Fig. 8. Late-heavy profile emerges; early layers drained, late layers retained.

10 Pareto Frontier — Params vs. BLEU

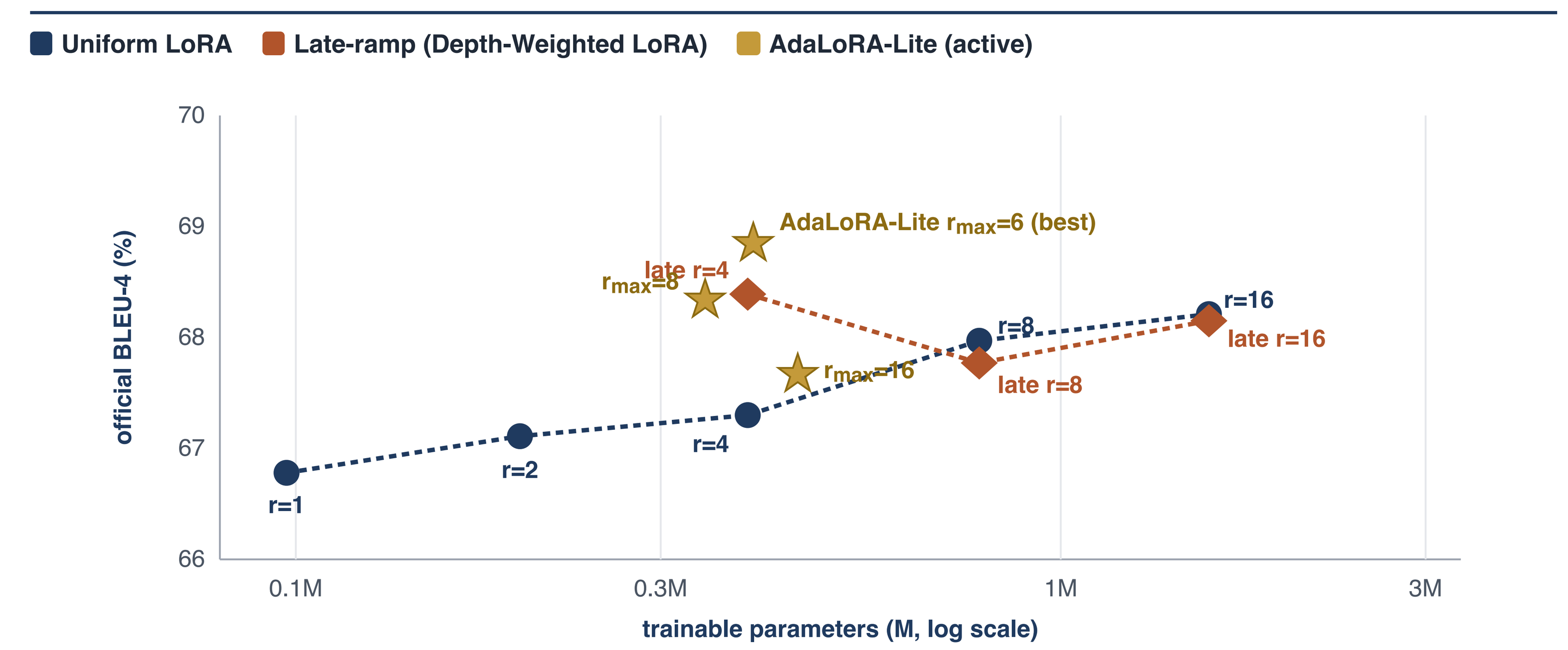

Fig. 9. AdaLoRA-Lite r_{max}=6 beats uniform r=16 with 4× fewer params.

Lesson: a small over-rank pruned gently > a large over-rank pruned aggressively.

QUESTION 3

Can one served model handle multiple tasks via routed LoRAs?

Independent of Q1 and Q2. Experts here are stock uniform r=4 LoRAs.

11 Extension #3 — Mixture-of-LoRA Experts (MoLE)

- Per-layer router picks **top-1** of N=3 frozen task LoRAs (E2E, DART, WebNLG) per token.
- Only the router trains: one `Linear(d→N)` per layer ≈ **0.07M** params.
- **Note:** experts are stock uniform r=4 LoRAs — *not* the specialised adapters from Q1–Q2 above. MoLE is a separate routing experiment.

$$\Delta h(x) = \text{scaling} \cdot \text{gate}_k(x) \cdot B_k(A_k x), \quad k = \operatorname{argmax}_j \text{gate}_j(x)$$

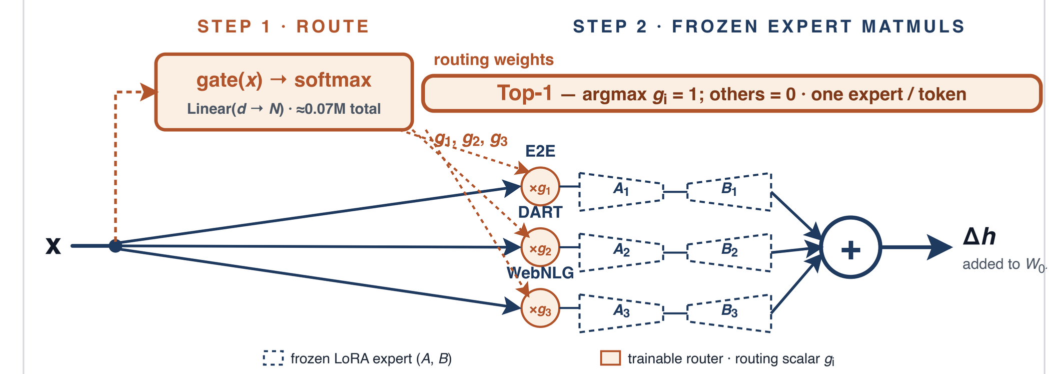

Fig. 10. Top-1 router picks one expert per token.

12 MoLE Results: Cross-Task Generalization

Method	Params	E2E BLEU	DART BLEU	WebNLG BLEU
E2E LoRA	0.39M	**67.30**	28.53	10.58
DART LoRA	0.39M	60.54	**47.15**	50.44
WebNLG LoRA	0.39M	39.75	33.41	**55.23**
MoLE (3 + router)	≈1.24M	67.76	45.35	51.75

MoLE total ≈1.24M = 3×0.39M frozen experts + 0.07M trainable router.

- Single-task LoRAs **fall apart off-domain** (E2E→DART drops ~39 BLEU). MoLE stays strong everywhere — **one** checkpoint.
- Matches/beats the E2E specialist on its home turf; trades a little on others.

Fig. 11. Bold = best in column. Each column tests on that task's own held-out split.

13 Conclusion & Future Work

- **Paper LoRA remains higher.** AdaLoRA-Lite r_{max}=6 beats paper full FT and roughly matches AdapterL at 0.39M active params.
- **Allocation ≈ raw rank.** Late-ramp r=4 beats uniform r=8 with half the params.
- **Learned ≈ hand-designed.** AdaLoRA-Lite recovers a late-heavy profile automatically.
- **MoLE.** One served model for E2E, DART, WebNLG via a tiny router.

Future work

- **Scale to larger backbones.** Move beyond GPT-2 Medium to GPT-2 Large and Llama-2/3 7B-class models.
- **MoLE + adaptive rank.** Combine mixture-of-LoRA routing with AdaLoRA-Lite–style learned rank budgets.
- **Orthogonality regularizer.** Add AdaLoRA-style SVD orthogonality penalties on projection factors (P, Q) to AdaLoRA-Lite and measure interference vs. pruning dynamics.
- **Key and Output matrices.** Experiment with applying LoRA and AdaLoRA-Lite to the Key and Output transformer projection matrices in addition to the Q/V matrices.

14 References

[1] Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., & Chen, W. (2021). *LoRA: Low-Rank Adaptation of Large Language Models.* arXiv:2106.09685.
[2] Zhang, Q. et al. (2023). *AdaLoRA: Adaptive Budget Allocation for Parameter-Efficient Fine-Tuning.* ICLR 2023 / arXiv:2303.10512.
[3] Novikova, J., Dušek, O., & Rieser, V. (2017). *The E2E Dataset: New Challenges for End-to-End Generation.* SIGDIAL 2017.
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[5] Li, X. L. & Liang, P. (2021). *Prefix-Tuning: Optimizing Continuous Prompts for Generation.* ACL 2021.
[6] Wu, X. et al. (2024). *Mixture-of-LoRA-Experts: Leveraging the Power of Fine-Tuned Models.* ICLR 2024 / arXiv:2404.13628.